

CO3-AT3-Assignment-2

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QUESTIONS

Q1 
Multithreaded Number
Processing
25 marks

Q2 
Railway Reservation 
Synchronization and Exception
Handling
25 marks

Q3 
Printer Queue 
Inter-Thread
Communication
25 marks

Q4 
Online Login System 
Multiple Exceptions
25 marks

4/4 answered

Q1 Multithreaded Number Processing

25 marks

Problem Statement:

Develop a Java program that creates multiple threads to perform different operations on a set of numbers.

Requirements:

Create separate threads to calculate the sum average and maximum value.

Use the Thread class.

Execute the threads concurrently.

Display the result produced by each thread

Your Code

```
1 public class main {
2     public static void main(String[] args) {
3         int[] a = {10, 20, 30, 40, 50};
4         Sum t1 = new Sum(a);
5         Average t2 = new Average(a);
6         Maximum t3 = new Maximum(a);
7         t1.start();
8         t2.start();
9         t3.start();
10    }
11 }
12 class Sum extends Thread {
13     int[] a;
14     Sum(int[] a) {
15         this.a = a;
16     }
17     public void run() {
18         int sum = 0;
19         for (int x : a) {
20             sum += x;
21         }
22     }
23 }
```

Input

stdin input

Output

```
Maximum = 50
Sum = 150
Average = 30.0
```

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QUESTIONS

Q1

Multithreaded Number Processing
25 marks



Q2

Railway Reservation ♦ Synchronization and Exception Handling
25 marks



Q3

Printer Queue ♦ Inter-Thread Communication
25 marks



Q4

Online Login System ♦ Multiple Exceptions
25 marks



4/4 answered

Q2 Railway Reservation ♦ Synchronization and Exception Handling

25 marks

Problem Statement:

Develop a Java program to simulate railway ticket reservation where multiple passengers attempt to reserve tickets concurrently.

Requirements:

Create multiple passenger threads.

Maintain a shared number of available tickets.

Synchronize the reservation method.

Create a user-defined exception for attempting to book unavailable tickets.

Handle the exception appropriately.

Your Code

```
1 public class main {
2     public static void main(String[] args) {
3         Railway r = new Railway(3);
4         Thread t1 = new Passenger(r, "Passenger 1");
5         Thread t2 = new Passenger(r, "Passenger 2");
6         Thread t3 = new Passenger(r, "Passenger 3");
7         Thread t4 = new Passenger(r, "Passenger 4");
8         Thread t5 = new Passenger(r, "Passenger 5");
9
10        t1.start();
11        t2.start();
12        t3.start();
13        t4.start();
14        t5.start();
15    }
16 }
17 class Passenger extends Thread {
18     private Railway r;
19     private String name;
```

Input

stdin input

Output

```
Passenger 1 booked a ticket.
Passenger 5 booked a ticket.
Passenger 4 booked a ticket.
Passenger 2: No tickets
available.
Passenger 3: No tickets
```

QUESTIONS

Q1
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25 marks

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Q3
Printer Queue & Inter-Thread Communication
25 marks

Q4
Online Login System
Multiple Exceptions
25 marks

4/4 answered

Q3 Printer Queue & Inter-Thread Communication

25 marks

Problem Statement:

Develop a Java program to simulate a shared printer used by multiple users. A user thread submits a document while a printer thread processes it.

Requirements:

Create producer and consumer threads.

Use a shared document queue/buffer.

Implement wait() and notify().

Ensure proper communication between the threads.

Display submission and printing status.

Your Code

```
1 public class Main {
2     public static void main(String[] args) {
3         Printer p = new Printer();
4         new User(p).start();
5         new Print(p).start();
6     }
7 }
8 class Printer {
9     private String doc;
10    private boolean ready = false;
11    synchronized void submit(String d) throws Exception {
12        while (!ready) wait();
13        doc = d;
14        ready = true;
15        System.out.println("Submitted: " + d);
16        notifyAll();
17    }
18    synchronized void print() throws Exception {
19        while (!ready) wait();
20        System.out.println("Printing: " + doc);
21        ready = false;
22        notifyAll();
23    }
24 }
```

Input

stdin input

Output

```
Submitted: File.pdf
Printing: File.pdf
```

Q1Multithreaded Number
Processing
25 marks**Q2**Railway Reservation
Synchronization and Exception
Handling
25 marks**Q3**Printer Queue
Inter-Thread
Communication
25 marks**Q4**Online Login System
Multiple Exceptions
25 marks

4/4 answered

Problem Statement:

Develop a Java program to simulate an online login system. The program should validate a username and password and perform operations that may generate different exceptions.

Requirements:

Accept login details.

Use try-catch-finally.

Handle at least two different built-in exceptions.

Display a message indicating whether the login operation was successful.

Ensure the finally block executes.

Your Code

```
1 import java.util.*;  
2 public class Main {  
3     public static void main(String[] args) {  
4         Scanner sc = new Scanner(System.in);  
5         try {  
6             System.out.print("Username: ");  
7             String user = sc.nextLine();  
8             System.out.print("Password: ");  
9             String pass = sc.nextLine();  
10            if (!user.equals("admin"))  
11                throw new IllegalArgumentException("Invalid Username");  
12            if (!pass.equals("1234"))  
13                throw new InputMismatchException("Invalid Password");  
14            System.out.println("Login Successful");  
15        }  
16        catch (IllegalArgumentException e) {  
17            System.out.println(e.getMessage());  
18        }  
19        catch (InputMismatchException e) {  
20            System.out.println(e.getMessage());  
21        }  
22        finally {  
23            System.out.println("Finally block executed");  
24            sc.close();  
25        }  
26    }  
27 }
```

Inputadmin
12345**Output**Username: Password: Invalid
Password
Finally block executed